exploratory-rmbl

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Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggridges)
```

Data

```
# Trail Creek Isotope
tc_isotope <- read_csv(here("data", "TC-isotopes_april-june.csv"), show_col_types = FALSE) |
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(site = case_when( # creating a new column of the site
    str_detect(sample, "UTC") ~ "utc",
    str_detect(sample, "TC")  ~ "tc",
```

```
    TRUE ~ NA_character_ )) |> # For names without UTC or TC
  filter(!is.na(site)) |>
  mutate(date = mdy(date)) # telling r the date format

# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_20260415_export20260630.csv"), show_col_types =
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_20260403_export20260630.csv"), show_col_types = FALSE)
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt,  # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE),  # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
```

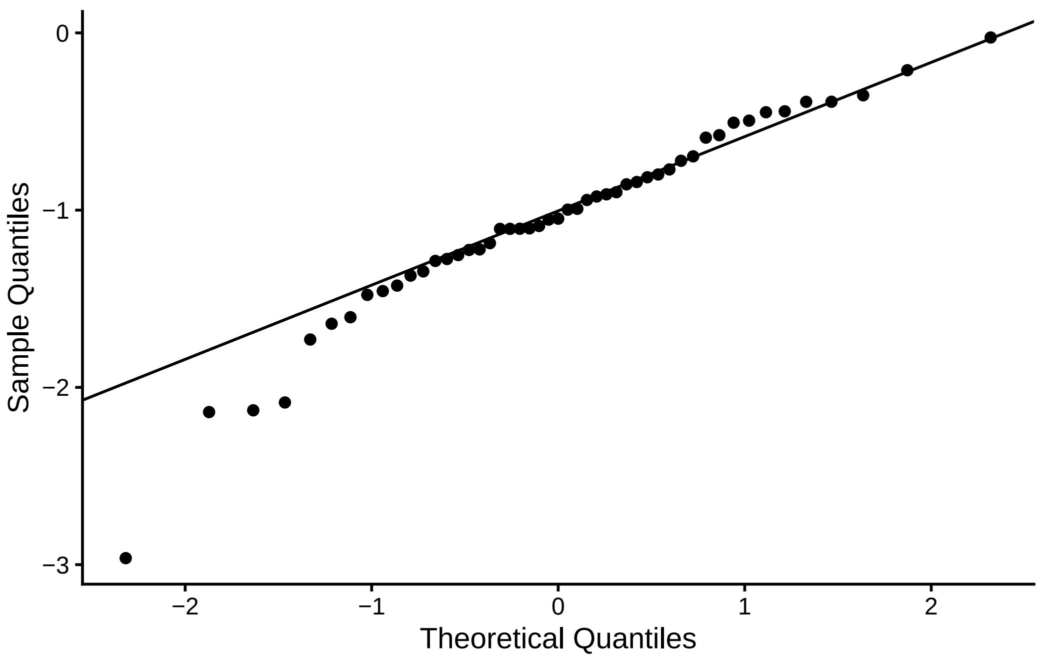
```
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site
```

```
tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data

isotope_data <- merge(tc_utc_data, tc_isotope, by = c("site", "date")) # merging that data wi
```

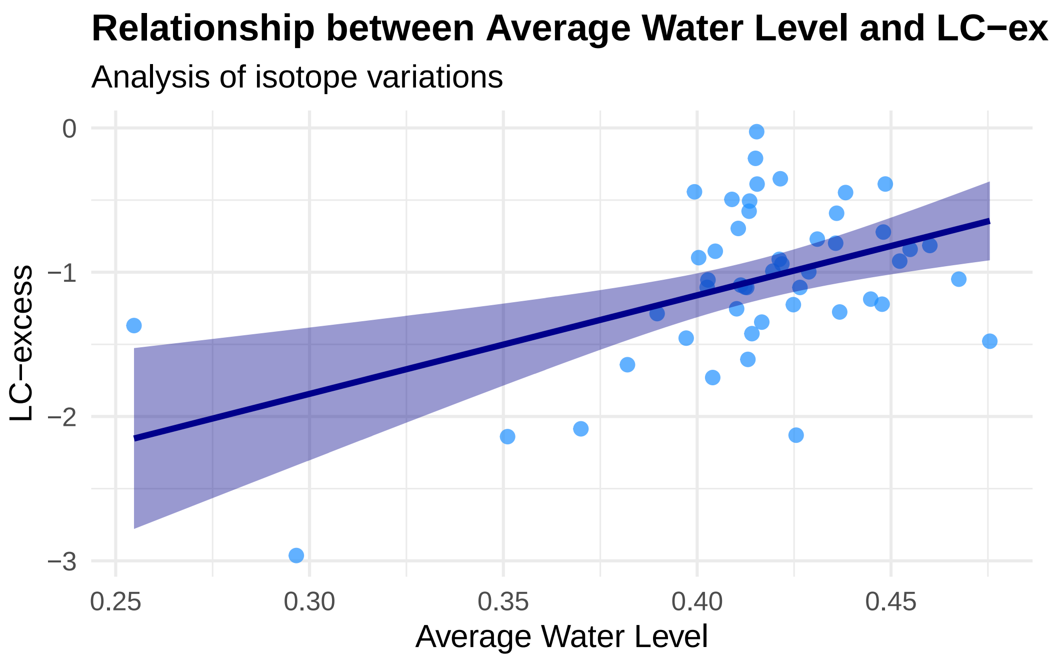
Exploratory Visuals

```
ggplot(isotope_data, aes(sample = lc_excess)) +
  stat_qq() +
  stat_qq_line(color = "black") +
  labs ( x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_classic()
```



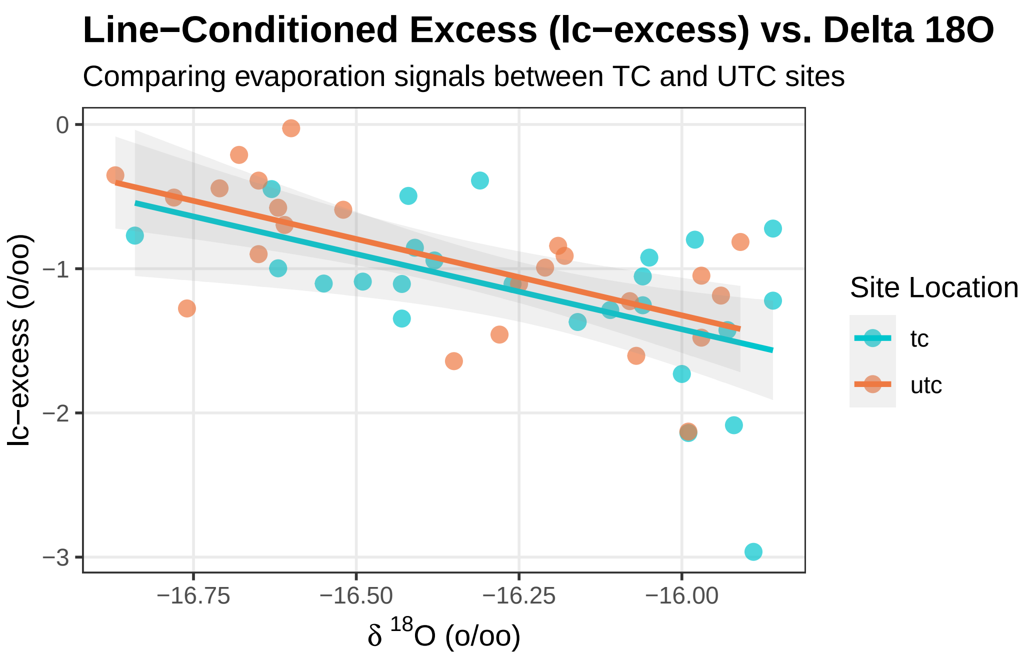

```
ggplot(data = isotope_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(color = "dodgerblue", size = 2, alpha = 0.7) +
  geom_smooth(method = "lm", color = "darkblue", fill = "darkblue", se = TRUE) +
  labs(
```

```
    title = "Relationship between Average Water Level and LC-excess",
    subtitle = "Analysis of isotope variations",
    x = "Average Water Level",
    y = "LC-excess"
  ) +
  theme_minimal(base_size = 12) +
   theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right")
```



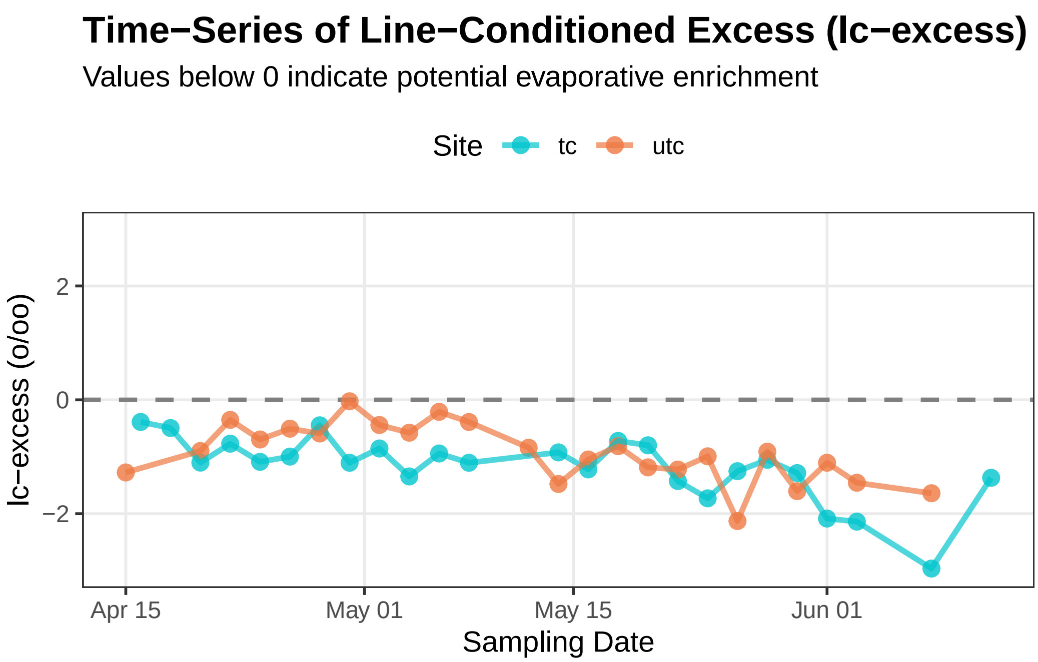

```
ggplot(data = isotope_data, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +
  scale_color_manual(values = c("tc" = "turquoise3", "utc" = "sienna2")) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
    subtitle = "Comparing evaporation signals between TC and UTC sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")), # Formats as  18O (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
```

```r
    panel.grid.minor = element_blank())
```



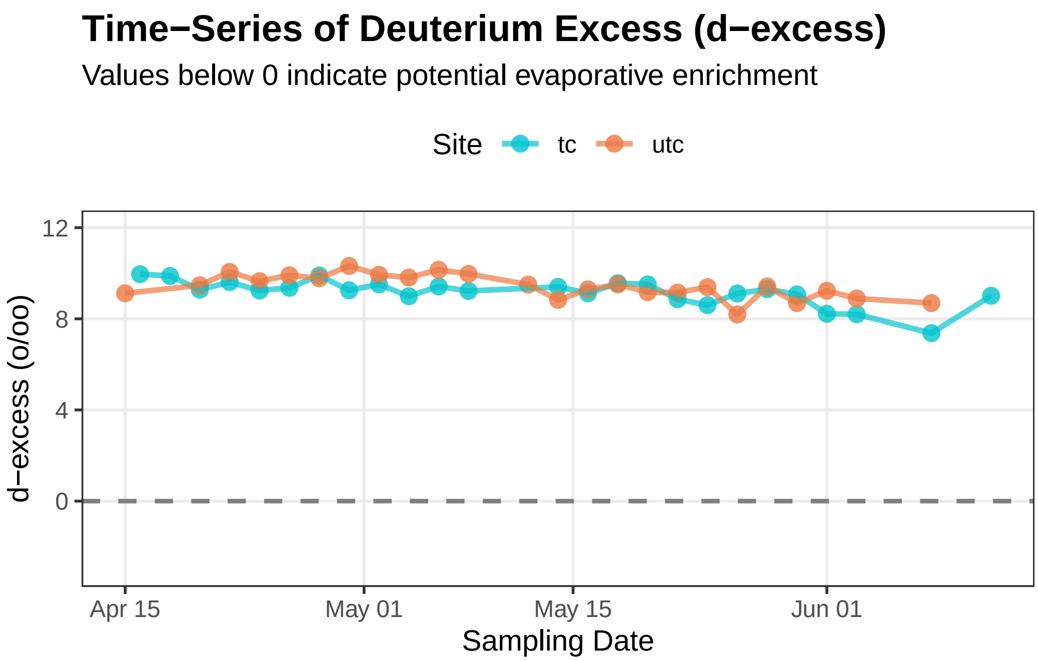

```r
ggplot(data = isotope_data,
       aes(x = date,
           y = lc_excess,
           color = site,
           group = site)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  scale_color_manual(values = c("tc" = "turquoise3",
                                "utc" = "sienna2")) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
```

```
  ylim(-3,3) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```

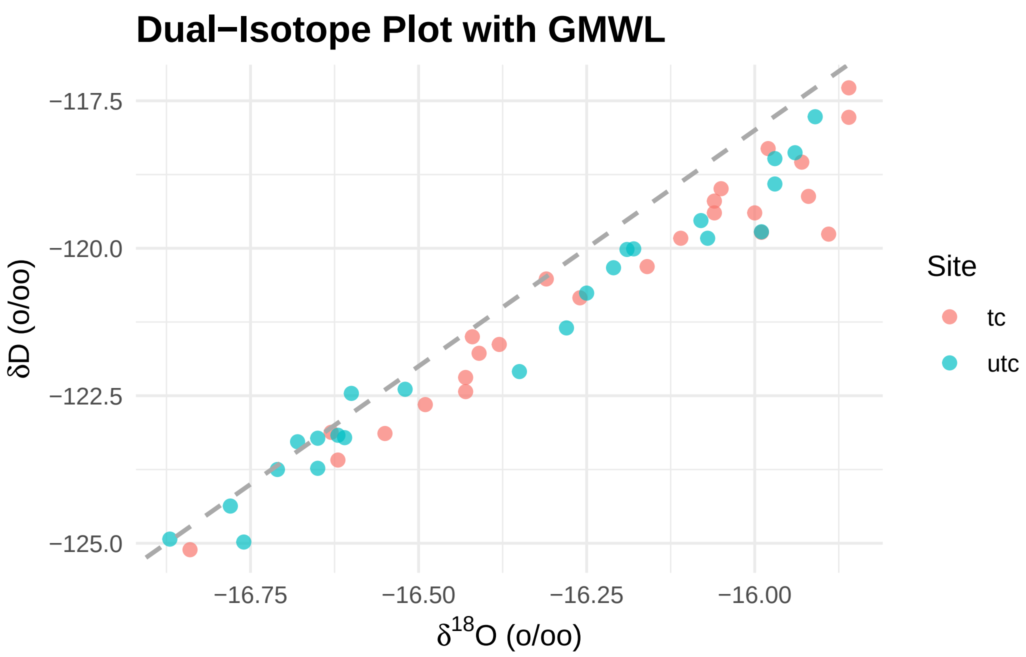



```
ggplot(data = isotope_data,
       aes(x = date,
           y = d_excess,
           color = site,
           group = site)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  scale_color_manual(values = c("tc" = "turquoise3",
                                "utc" = "sienna2")) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
```

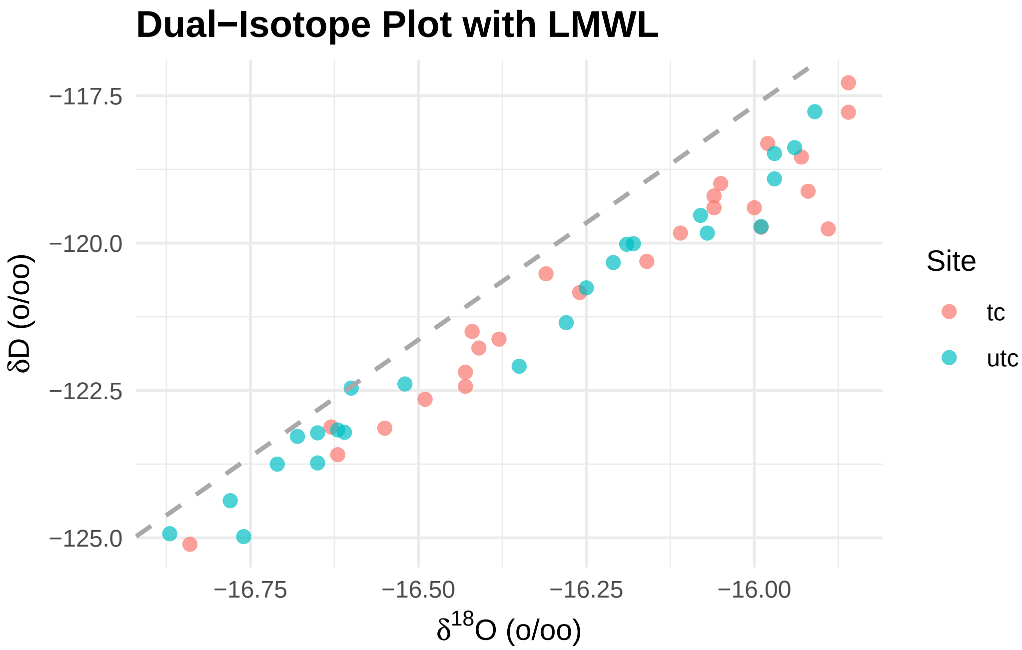
```
      subtitle = "Values below 0 indicate potential evaporative enrichment",
      x = "Sampling Date",
      y = "d-excess (\u2030)",
      color = "Site") +
  ylim(-3,12) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



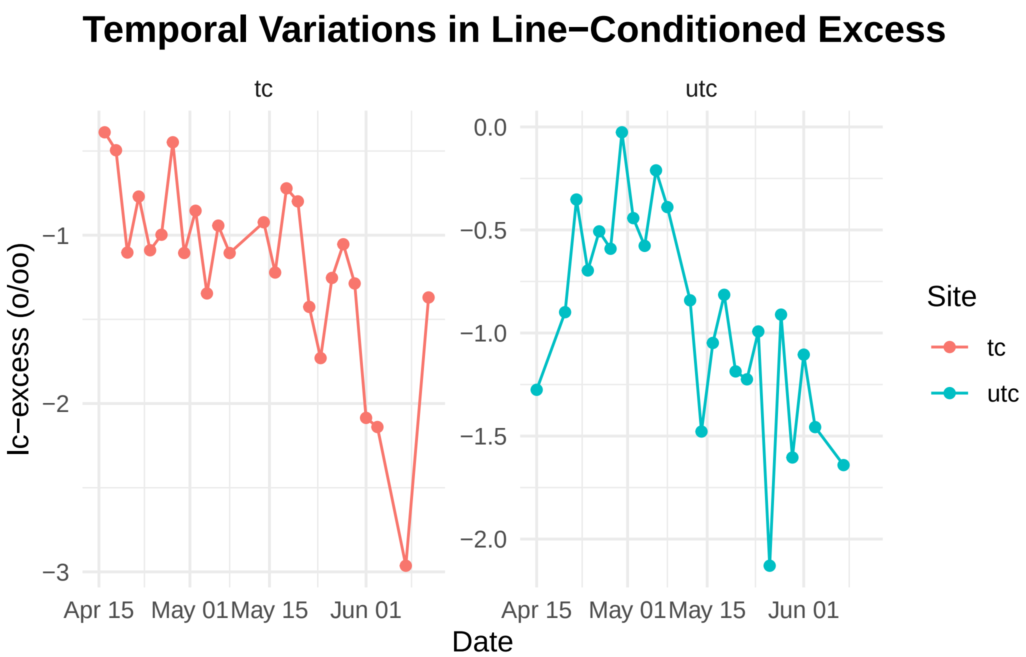

```
ggplot(isotope_data, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray", linewidth =
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



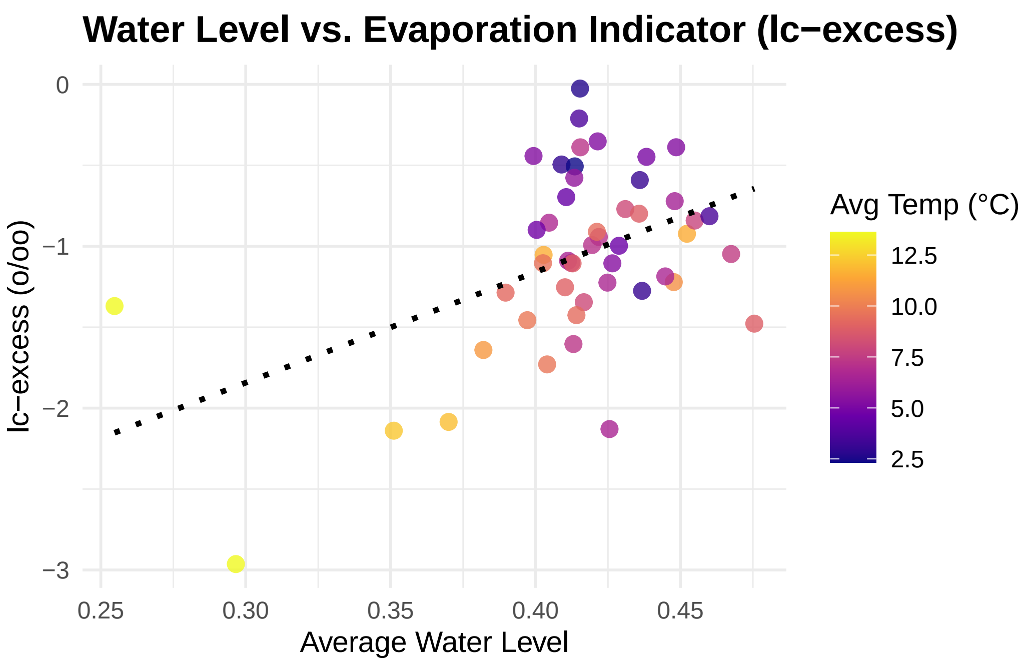

```r
ggplot(isotope_data, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Local Mean Water Line (LMWL: y = 7.94 x + 9.37 )
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewi
  labs(
    title = "Dual-Isotope Plot with LMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
   theme(plot.title = element_text(face = "bold", size = 14))
```



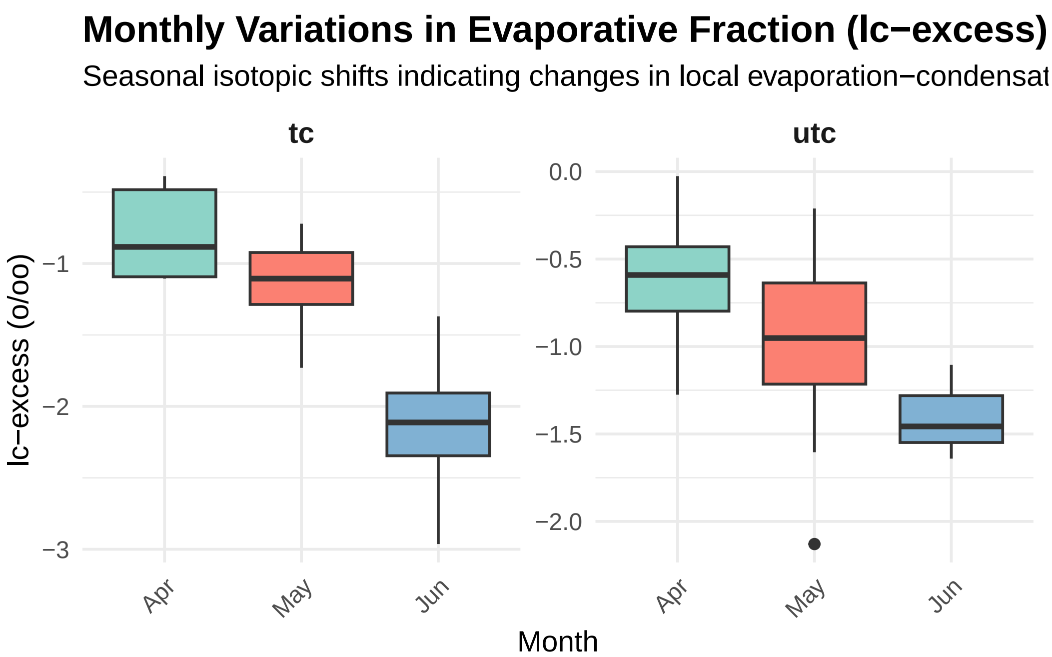

```r
ggplot(isotope_data, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") +  # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```




```
ggplot(isotope_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(aes(color = avg_temp), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", color = "black", linetype = "dotted", se = FALSE) +
  scale_color_viridis_c(option = "plasma") +  # Beautiful temperature gradient
  labs(
    title = "Water Level vs. Evaporation Indicator (lc-excess)",
    x = "Average Water Level",
    y = "lc-excess (‰)",
    color = "Avg Temp (°C)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```




```
ggplot(isotope_data, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y = lc_excess))
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~site, scales = "free_y") +
   scale_fill_manual(values = c("#8dd3c7", "#fb8072", "#80b1d3")) +
 labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-condensatio
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1)   # Tilt month labels if they overlap
  )
```


Monthly Variations in Evaporative Fraction (lc−excess)
Seasonal isotopic shifts indicating changes in local evaporation−condensat
tc
utc
lc−excess (o/oo)
−1
−2
−3
0.0
−0.5
−1.0
−1.5
−2.0
Apr
May
Jun
Apr
May
Jun
Month